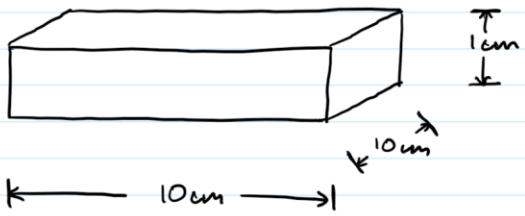
Consider an aluminium rectangular prism with the following dimensions.

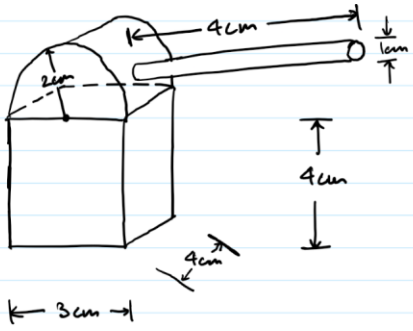


We know that aluminium has a density of $2.7\ g/cm^3$ while specialist alloys for fire resistant applications, like Alloy 5052-H32 has a density of $2.68\ g/m^3$. Since we are using this to select a motor with a suitable torque lets consider the denser pure alloy.

$$\rho = \frac{m}{V}$$

$$m = \rho V$$
$$= 2.7\,(10)^2$$
$$\therefore m = 270 \text{ grams.}$$

We know that this plate is going to also hold a turret which will have the following approximate dimensions.



$$V = (4\times3\times4) + \pi(4) + \frac{\pi(2)^2}{2}(4)$$
$$= 48 + 4\pi + 8\pi$$
$$\therefore V = 48 + 12\pi$$

Assuming we also use aluminium of density $2.7\ g/cm^3$, this turret will have a mass of

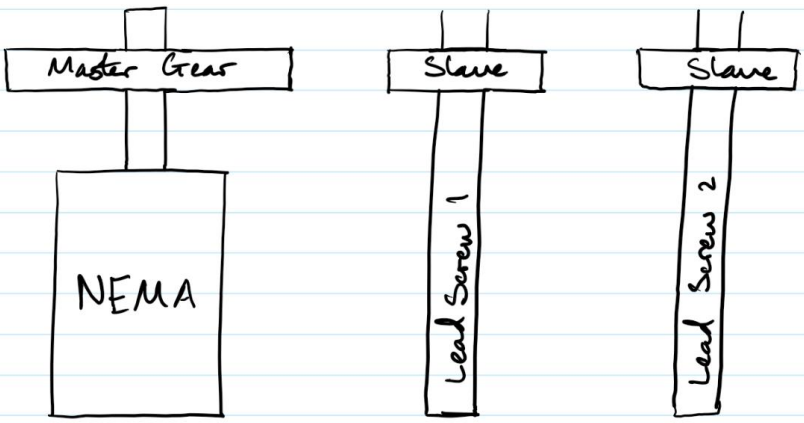
$$m = 2.7\,(48 + 12\pi)$$
$$m = 231.3876....$$

Additionally our turret will need a NEMA Stepper Motor, either a 14 or 17 Pancake, to pivot the turret gun vertically. This will have a weight between 120g to 134g. Taking the upper bound we will consider the motor weight to be 134g.

Hence our approximate total gantry mass of 635.39 grams.

From here we can calculate the gravitational load that will be applied on the motor.

$$F = mg = 0.63539(9.81) = 6.23317.... = 6.233 \text{ N.}$$



Given the above transmission system and assuming we use ACME trapezoidal lead screws with a major diameter of 8 mm and lead of 2 mm

$$T_{raise} = \frac{F d_m}{2}\left(\frac{L + \pi \mu d_m \sec\alpha}{\pi d_m - \mu L \sec\alpha}\right)$$

Coefficient of friction for leadscrew threads[11][12]

Screw material	Nut material			
	Steel	Bronze	Brass	Cast iron
Steel, dry	0.15–0.25	0.15–0.23	0.15–0.19	0.15–0.25
Steel, machine oil	0.11–0.17	0.10–0.16	0.10–0.15	0.11–0.17
Bronze	0.08–0.12	0.04–0.06	-	0.06–0.09

By definition we know that $\alpha = 15°$ for an ACME Trapezoidal lead screw and we can assume a coefficient of friction to be 0.25.

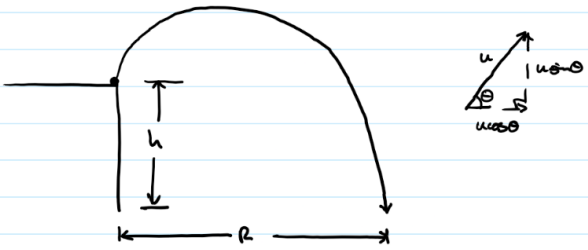
$$\therefore T_{raise} = \frac{(6.233)(0.008)}{2}\left(\frac{0.002 + \pi(0.25)(0.008)\sec 30°}{\pi(0.008) - (0.25)(0.002)\sec 30°}\right)$$

$$\therefore T_{raise} = 0.008614\ldots \text{ Nm}$$

$$\therefore T_{raise} = 0.0086 \text{ Nm}.$$

We can apply a safety margin by doubling this value. Thus we need the gantry motor to provide at least 0.0172 Nm of torque.

Before designing the turret we first need to identify which variables need to be maximised to ensure we maximise the robots extinguishing range. Maximising this range allows us to maintain our distance from the fire, thus allowing us to minimise heat shield costs and increase safety. Consider the following projectile motion model.



$\underset{\sim}{a}_y = -g$

$\underset{\sim}{v}_y = -gt + c$

when $t=0$, $\underset{\sim}{v}_y = u\sin\theta$

$\underset{\sim}{v}_y = -gt + u\sin\theta$

$\underset{\sim}{s}_y = \int -gt + u\sin\theta \, dt$

$= -\frac{gt^2}{2} + ut\sin\theta + c_1$

when $t=0$, $\underset{\sim}{s}_y = h$.

$\therefore \underset{\sim}{s}_y = \frac{-gt^2}{2} + ut\sin\theta + h$.

let $\underset{\sim}{s}_y = 0$ for the range of the projectile.

$0 = \frac{-gt^2}{2} + ut\sin\theta + h$

$t = \frac{-u\sin\theta \pm \sqrt{(u\sin\theta)^2 + 4\frac{g}{2}(h)}}{2}$

$t = \frac{u\sin\theta \mp \sqrt{u^2\sin^2\theta + 2gh}}{g}$

$\therefore t = \frac{u\sin\theta + \sqrt{u^2\sin^2\theta + 2gh}}{g}$ $\left(t>0,\ \sqrt{u^2\sin^2\theta+2gh} > u\sin\theta\right)$

$\therefore R = u\cos\theta\left(\frac{u\sin\theta + \sqrt{u^2\sin^2\theta+2gh}}{g}\right)$

For max height let $v_y = 0$.

$0 = -gt + u\sin\theta$

$t = \frac{u\sin\theta}{g}$

$\therefore H = -\frac{g\left(\frac{u\sin\theta}{g}\right)^2}{2} + \left(\frac{u\sin\theta}{g}\right)u\sin\theta + h$

$= -\frac{u^2\sin^2\theta}{2g} + \frac{u^2\sin^2\theta}{g} + h$

$= \frac{u^2\sin^2\theta}{g} + h$

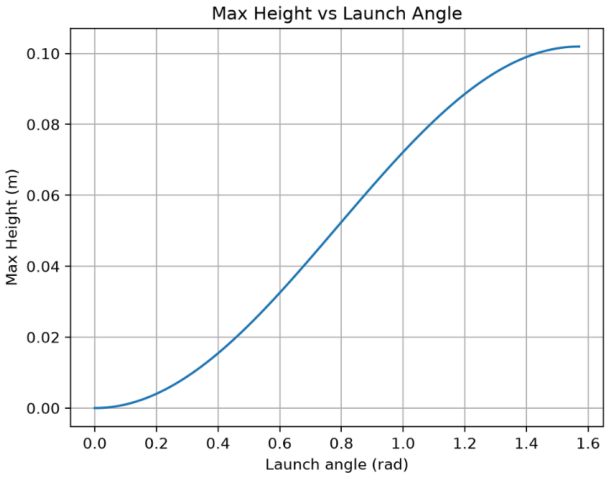
$\underset{\sim}{a}_x = 0$

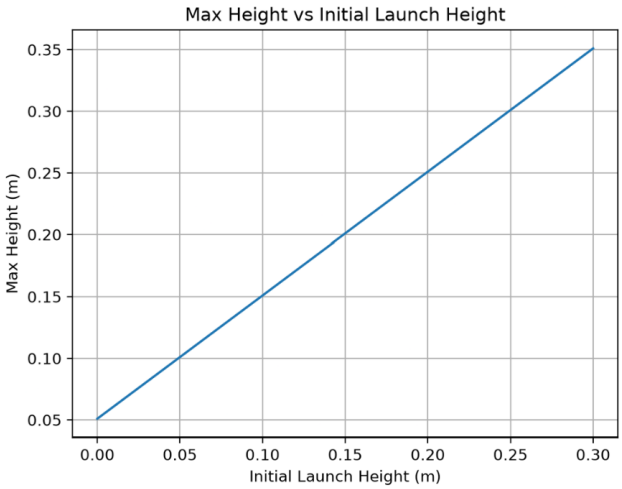
$\underset{\sim}{v}_x = 0 + c_1$

when $t=0$, $\underset{\sim}{v}_x = u\cos\theta$.

$\therefore \underset{\sim}{v}_x = 0 + u\cos\theta$

$\therefore s_x = ut\cos\theta$






```python
# _____ Max Height with Varying Theta _____
u = 1
h = 0
Pi = np.pi
g = 9.81

def maxHeight(theta):
    return ((u**2 * (np.sin(theta))**2) / g + h)

theta = np.linspace(0, Pi/2, 100000)

y = maxHeight(theta)

plt.plot(theta, y)
plt.grid(True)
plt.xlabel("Launch angle (rad)")
plt.ylabel("Max Height (m)")
plt.title("Max Height vs Launch Angle")
plt.show()
```

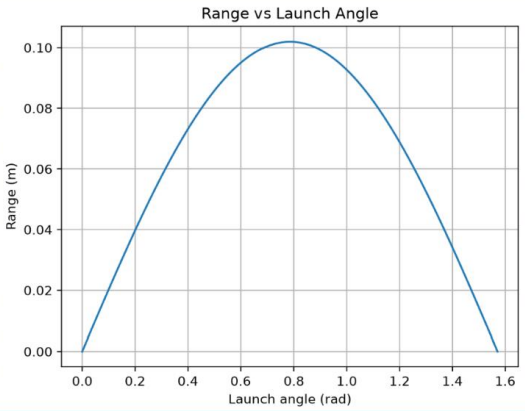
```python
# _____ Max Height with Varying Launch Height _____
u = 1
Pi = np.pi
theta = Pi/4
g = 9.81

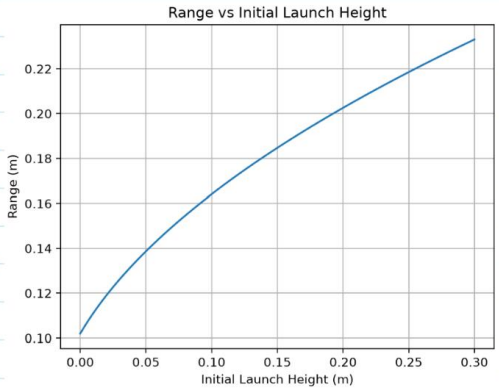
def maxHeight(h):
    return ((u**2 * (np.sin(theta))**2) / g + h)

h = np.linspace(0, 0.3, 100000)

y = maxHeight(h)

plt.plot(h, y)
plt.grid(True)
plt.xlabel("Initial Launch Height (m)")
plt.ylabel("Max Height (m)")
plt.title("Max Height vs Initial Launch Height")
plt.show()
```






```python
# _____ Range with Varying Theta _____
u = 1
h = 0
Pi = np.pi
g = 9.81

def range(theta):
    return (u * np.cos(theta)) * (u * np.sin(theta) + np.sqrt(u**2 * (np.sin(theta))**2 + 2*g*h)) / g

theta = np.linspace(0, Pi/2, 100000)

y = range(theta)

plt.plot(theta, y)
plt.grid(True)
plt.xlabel("Launch angle (rad)")
plt.ylabel("Range (m)")
plt.title("Range vs Launch Angle")
plt.show()
```

```python
# _____ Range with Varying Launch Height _____
u = 1
Pi = np.pi
theta = Pi/4
g = 9.81

def Range(h):
    return (u * np.cos(theta)) * (u * np.sin(theta) + np.sqrt(u**2 * (np.sin(theta))**2 + 2*g*h)) / g

h = np.linspace(0, 0.3, 100000)

y = Range(h)

plt.plot(h, y)
plt.grid(True)
plt.xlabel("Initial Launch Height (m)")
plt.ylabel("Range (m)")
plt.title("Range vs Initial Launch Height")
plt.show()
```

It is clear from the above plots that increasing the gantry height will result in the largest extension of our robots extinguishing range. Thus validating the need to have a gantry raising system.

For the turret nozzle design we need to maximise water velocity to increase the the extinguishing range. By the venturi effect we know that

$$P_1 + \frac{\rho v_1^2}{2} + \gamma \cancel{z_1}^{0} = P_2 + \frac{\rho v_2^2}{2} + \gamma \cancel{z_2}^{0} + \Delta P_{loss}$$

$$P_1 - P_2 = \frac{\rho v_2^2}{2} - \frac{\rho v_1^2}{2} + \frac{\rho K v_2^2}{2}$$

$$P_1 - P_2 = \frac{\rho}{2}\left(v_2^2(1+K) - v_1^2\right)$$

This gives us system resistance curve.
By the continuity equation we know

$$Q = A_1 v_1 = A_2 v_2$$

$$v_2 = \frac{Q}{A_2}$$

Our selected pump has a pump curve of

$$P_1(Q) = 60000 - 1.54 \times 10^{13} Q^2 \quad (Q \text{ in } m^3/s). \text{ (gauge pressure)}$$

We know that $P_2 = P_{atm}$ when it exits the nozzle

$$\therefore \quad P_1(Q) = P_{atm} + \frac{\rho}{2}\left((1+K)\left(\frac{Q}{A_2}\right)^2 - \frac{Q^2}{A_1^2}\right)$$

$$60000 - 1.54\times10^{13} Q^2 = \frac{\rho}{2}\left((1+K)\left(\frac{Q}{\pi r^2}\right)^2 - \frac{Q^2}{A_1^2}\right)$$

$$60000 - (1.54\times10^{13}) Q^2 = \frac{\rho(1+K)}{2\pi^2 r^4} Q^2 - \frac{Q^2\rho}{2A_1^2}$$

$$60000 = Q^2\left(\frac{\rho(1+K)}{2\pi^2 r^4} + 1.54\times10^{13} - \frac{\rho}{2A_1^2}\right)$$

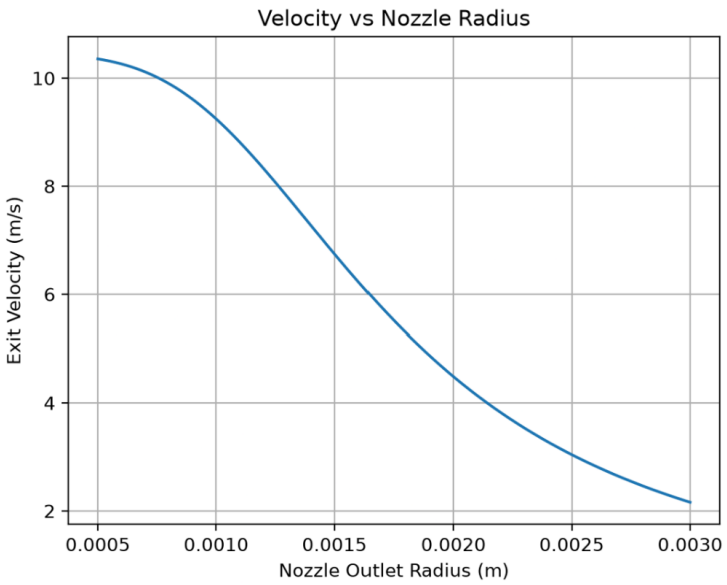
$$Q = \sqrt{\frac{60000}{\frac{\rho}{2}\left(\frac{1+K}{\pi^2 r^4} - \frac{1}{A_1^2}\right) + (1.54\times10^{13})}}$$

Assuming a K value of 0.1 as per Crane's Technical Paper 410

$$Q = \sqrt{\frac{60000}{\frac{1000}{2}\left(\frac{1+0.1}{\pi^2 r^4} - \frac{1}{A_1^2}\right) + (1.54\times10^{13})}}$$

$$v_2 = \frac{Q}{A_2}, \quad A_1 = \pi(0.006)^2$$

$$v_2 = \frac{\sqrt{\frac{60000}{500\left(\frac{1.1}{\pi^2 r^4} - \frac{1}{\pi(0.006)^4}\right) + (1.54\times10^{13})}}}{\pi r^2}$$




```python
import numpy as np
import matplotlib.pyplot as plt

# _____ Constants _____
rho = 1000
r_in = 0.003
K = 0.1
Pi = np.pi
P0 = 60000
k_pump = 1.54e13

r = np.linspace(0.0005, r_in, 10000)

def vel(r):
    R = (rho / 2) * (((1+K) / (np.pi**2*r**4)) - (1 / (Pi * (r_in)**2)))
    Q = np.sqrt(P0 / (k_pump + R))
    A = Pi * (r)**2
    return Q/A

y = vel(r)

plt.plot(r, y)
plt.grid(True)
plt.xlabel("Nozzle Outlet Radius (m)")
plt.ylabel("Exit Velocity (m/s)")
plt.title("Velocity vs Nozzle Radius")
plt.show()
```

Considering an inlet radius of 3 mm we should aim for a nozzle outlet of radius 1 mm.

